

LPII Viva Questions & Answers

Laboratory Practice II – Third Year Engineering
Comprehensive Topic-wise Q&A for Oral Examinations

- Graph Traversal – DFS & BFS
 - A* (A Star) Algorithm
 - Greedy Algorithm – Prim's & Kruskal's (MST)
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Graph Traversal – DFS & BFS

Q1. What is a graph?

Ans: A graph is a data structure consisting of vertices (nodes) and edges connecting them.

Q2. What is an undirected graph?

Ans: An undirected graph is a graph where edges do not have direction, so traversal is possible in both ways.

Q3. What is graph traversal?

Ans: Graph traversal means visiting all vertices of a graph systematically.

Q4. What is Depth First Search (DFS)?

Ans: DFS is a traversal algorithm that explores a branch completely before backtracking.

Q5. What is Breadth First Search (BFS)?

Ans: BFS is a traversal algorithm that visits all neighboring vertices level by level.

Q6. Which data structure is used in DFS?

Ans: DFS mainly uses a stack or recursion.

Q7. Which data structure is used in BFS?

Ans: BFS uses a queue.

Q8. Why is recursion used in DFS?

Ans: Recursion automatically stores function calls in the call stack, making traversal easier.

Q9. What is backtracking in DFS?

Ans: Backtracking means returning to the previous vertex when no unvisited neighbor exists.

Q10. What is the time complexity of DFS?

Ans: $O(V + E)$, where V = vertices and E = edges.

Q11. What is the time complexity of BFS?

Ans: $O(V + E)$, where V = vertices and E = edges.

Q12. What does V and E represent?

Ans: V represents vertices and E represents edges in a graph.

Q13. What is an adjacency matrix?

Ans: An adjacency matrix is a 2D array used to represent graph connections.

Q14. What is an adjacency list?

Ans: An adjacency list stores connected vertices in list form for each node.

Q15. Which traversal is better for finding the shortest path?

Ans: BFS is better for finding the shortest path in an unweighted graph.

Q16. Which traversal uses more memory?

Ans: BFS generally uses more memory because it stores all neighboring nodes in a queue.

Q17. Can DFS be implemented iteratively?

Ans: Yes, DFS can be implemented using a stack without recursion.

Q18. What is the starting vertex in graph traversal?

Ans: It is the vertex from which traversal begins.

Q19. Why do we use a visited array?

Ans: A visited array prevents revisiting the same vertex and avoids infinite loops.

Q20. What happens if visited nodes are not marked?

Ans: The algorithm may enter an infinite loop in cyclic graphs.

Q21. What is a connected graph?

Ans: A connected graph is a graph where every vertex is reachable from any other vertex.

Q22. Can BFS and DFS work on trees?

Ans: Yes, both algorithms can traverse tree structures.

Q23. What is the main difference between BFS and DFS?

Ans: BFS explores level by level, while DFS explores depth-wise.

Q24. Which traversal is recursive by nature?

Ans: DFS is recursive by nature.

Q25. Give one application of DFS.

Ans: DFS is used in cycle detection and topological sorting.

Q26. Give one application of BFS.

Ans: BFS is used in shortest path finding and web crawling.

Q27. What is a queue?

Ans: A queue follows FIFO (First In First Out) order.

Q28. What is a stack?

Ans: A stack follows LIFO (Last In First Out) order.

Q29. Can DFS guarantee the shortest path?

Ans: No, DFS does not guarantee the shortest path.

Q30. Can BFS guarantee the shortest path in an unweighted graph?

Ans: Yes, BFS guarantees the shortest path in an unweighted graph.

Q31. What is a recursive function?

Ans: A recursive function is a function that calls itself repeatedly.

Q32. What is the base condition in recursion?

Ans: The base condition stops recursive calls and prevents infinite recursion.

Q33. What is a cyclic graph?

Ans: A cyclic graph contains at least one cycle.

Q34. Why is DFS called depth-first?

Ans: Because it explores as deep as possible before moving to another branch.

Q35. Why is BFS called breadth-first?

Ans: Because it explores all neighboring nodes first before going deeper.

Q36. Which traversal is suitable for social networking applications?

Ans: BFS is suitable because it explores nearest connections first.

Q37. Which traversal is used in maze solving?

Ans: DFS is commonly used in maze solving.

Q38. What is spanning tree traversal?

Ans: It is visiting all vertices of a graph without forming cycles.

Q39. Can graphs have disconnected components?

Ans: Yes, some graphs may contain disconnected components.

Q40. How do you traverse a disconnected graph?

Ans: Traversal is started separately from each unvisited vertex.

A* (A Star) Algorithm

Q1. What is the A* Algorithm?

Ans: A* is a search algorithm used to find the shortest and optimal path between nodes.

Q2. Where is the A* algorithm used?

Ans: It is used in games, robotics, maps, and pathfinding applications.

Q3. What type of algorithm is A*?

Ans: A* is an informed search algorithm.

Q4. Why is A* called an informed search algorithm?

Ans: Because it uses heuristic information to guide the search.

Q5. What is the main objective of A*?

Ans: The main objective is to find the minimum cost path.

Q6. What is the evaluation function of A*?

Ans: $f(n) = g(n) + h(n)$

Q7. What does $f(n)$ represent?

Ans: $f(n)$ represents the total estimated cost of the path.

Q8. What does $g(n)$ represent?

Ans: $g(n)$ represents the actual cost from the start node to the current node.

Q9. What does $h(n)$ represent?

Ans: $h(n)$ represents the heuristic estimated cost from the current node to the goal node.

Q10. What is a heuristic function?

Ans: A heuristic function estimates the distance or cost to reach the goal.

Q11. Why is heuristic important in A*?

Ans: It helps reduce unnecessary searching and improves efficiency.

Q12. What happens if $h(n) = 0$?

Ans: A* behaves like Dijkstra's algorithm.

Q13. Which data structure is commonly used in A*?

Ans: Priority Queue is commonly used.

Q14. What is an open list in A*?

Ans: The open list contains nodes that are yet to be explored.

Q15. What is a closed list in A*?

Ans: The closed list contains nodes that are already explored.

Q16. What is the time complexity of A*?

Ans: The worst-case time complexity is exponential.

Q17. What is pathfinding?

Ans: Pathfinding means finding the shortest route between two points.

Q18. What is the advantage of A* over BFS?

Ans: A* is faster because it uses heuristics.

Q19. What is the advantage of A* over DFS?

Ans: A* can find the optimal shortest path.

Q20. Is A* complete?

Ans: Yes, A* is complete if the heuristic is admissible.

Q21. Is A* optimal?

Ans: Yes, A* gives the optimal solution when the heuristic is admissible.

Q22. What is an admissible heuristic?

Ans: A heuristic that never overestimates the actual cost.

Q23. Give an example of heuristic in games.

Ans: Manhattan distance in puzzle games.

Q24. What is Manhattan distance?

Ans: $|x_1 - x_2| + |y_1 - y_2|$

Q25. What is Euclidean distance?

Ans: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Q26. Which heuristic is commonly used in grid games?

Ans: Manhattan distance heuristic.

Q27. What are nodes in A*?

Ans: Nodes represent states or positions in the search space.

Q28. What is the goal node?

Ans: The destination node to be reached.

Q29. What is the start node?

Ans: The node from which the search begins.

Q30. Can A* work on weighted graphs?

Ans: Yes, A* works efficiently on weighted graphs.

Q31. What is the role of cost in A*?

Ans: Cost determines the best path among alternatives.

Q32. What is greedy best-first search?

Ans: It selects nodes only using heuristic values.

Q33. Difference between Greedy Search and A*?

Ans: Greedy uses only $h(n)$, while A* uses both $g(n)$ and $h(n)$.

Q34. What is backtracking in A*?

Ans: Tracing back the final optimal path from goal to start.

Q35. Why is A* widely used in games?

Ans: Because it efficiently finds shortest paths for characters and objects.

Q36. What is state space search?

Ans: Searching through all possible states to reach the goal.

Q37. Can A* fail to find a solution?

Ans: Yes, if no valid path exists.

Q38. What are the limitations of A*?

Ans: It consumes more memory for large graphs.

Q39. What is priority queue?

Ans: A queue where elements are processed based on priority.

Q40. Give real-life applications of A* algorithm.

Ans: Google Maps, robot navigation, video games, and network routing.

Greedy Algorithm – Prim's & Kruskal's (MST)

Q1. What is a Greedy Algorithm?

Ans: A greedy algorithm makes the best possible choice at each step to reach an optimal solution.

Q2. What is a Minimum Spanning Tree (MST)?

Ans: An MST is a tree connecting all vertices with minimum total edge cost and no cycles.

Q3. What is a spanning tree?

Ans: A spanning tree is a subgraph that connects all vertices without cycles.

Q4. What is the main objective of MST algorithms?

Ans: To connect all vertices with minimum total weight.

Q5. Which algorithms are used to find MST?

Ans: Prim's Algorithm and Kruskal's Algorithm.

Q6. What type of graph is required for MST?

Ans: A connected and weighted undirected graph.

Q7. What is Prim's Algorithm?

Ans: Prim's Algorithm builds the MST by selecting minimum weight edges from a starting vertex.

Q8. What is Kruskal's Algorithm?

Ans: Kruskal's Algorithm builds MST by selecting the smallest edges without forming cycles.

Q9. Which strategy is used in Prim's Algorithm?

Ans: Greedy strategy.

Q10. Which strategy is used in Kruskal's Algorithm?

Ans: Greedy strategy.

Q11. What data structure is commonly used in Prim's Algorithm?

Ans: Priority Queue or Min Heap.

Q12. What data structure is used in Kruskal's Algorithm?

Ans: Disjoint Set (Union-Find).

Q13. What is the time complexity of Prim's Algorithm?

Ans: $O(E \log V)$

Q14. What is the time complexity of Kruskal's Algorithm?

Ans: $O(E \log E)$

Q15. What is a weighted graph?

Ans: A graph in which edges have associated weights or costs.

Q16. Why is cycle detection important in Kruskal's Algorithm?

Ans: To avoid forming loops in the MST.

Q17. What is a cycle in a graph?

Ans: A cycle is a path that starts and ends at the same vertex.

Q18. How does Prim's Algorithm start?

Ans: It starts from any arbitrary vertex.

Q19. How does Kruskal's Algorithm start?

Ans: It starts by sorting all edges in increasing order.

Q20. What is the difference between Prim's and Kruskal's Algorithm?

Ans: Prim's works on vertices, while Kruskal's works on edges.

Q21. Which algorithm is better for dense graphs?

Ans: Prim's Algorithm.

Q22. Which algorithm is better for sparse graphs?

Ans: Kruskal's Algorithm.

Q23. Can MST exist for disconnected graphs?

Ans: No, MST exists only for connected graphs.

Q24. How many edges are present in MST with V vertices?

Ans: $V - 1$

Q25. Why are greedy methods called greedy?

Ans: Because they choose the locally optimal solution at each step.

Q26. What is the advantage of Prim's Algorithm?

Ans: It efficiently handles dense graphs.

Q27. What is the advantage of Kruskal's Algorithm?

Ans: It is simple and efficient for sparse graphs.

Q28. What is Union-Find?

Ans: A data structure used to detect cycles in Kruskal's Algorithm.

Q29. What is edge relaxation?

Ans: Updating the minimum cost edge during processing.

Q30. Can MST have negative edge weights?

Ans: Yes, MST algorithms work with negative weights.

Q31. Is MST unique for every graph?

Ans: No, multiple MSTs can exist if edge weights are equal.

Q32. What happens if all edge weights are distinct?

Ans: The MST will be unique.

Q33. What are real-life applications of MST?

Ans: Network design, road construction, and cable connections.

Q34. Why is sorting needed in Kruskal's Algorithm?

Ans: To select minimum weight edges first.

Q35. What is the output of Prim's and Kruskal's Algorithm?

Ans: A Minimum Spanning Tree with minimum total cost.

Backtracking & Branch and Bound – N-Queens & Graph Coloring

Q1. What is a Constraint Satisfaction Problem (CSP)?

Ans: A CSP is a problem where values are assigned to variables while satisfying given constraints.

Q2. What are the main components of CSP?

Ans: Variables, domains, and constraints.

Q3. What is Backtracking?

Ans: Backtracking is a recursive technique that tries solutions and removes invalid choices.

Q4. What is Branch and Bound?

Ans: Branch and Bound is an optimization technique that reduces the search space using bounds.

Q5. What is the N-Queens Problem?

Ans: The N-Queens problem places N queens on a chessboard so that no two queens attack each other.

Q6. What are the constraints in the N-Queens problem?

Ans: No two queens should be in the same row, column, or diagonal.

Q7. What is Graph Coloring?

Ans: Graph coloring assigns colors to vertices such that adjacent vertices have different colors.

Q8. What is the objective of Graph Coloring?

Ans: To minimize the number of colors while satisfying constraints.

Q9. Which algorithm is commonly used for N-Queens?

Ans: Backtracking algorithm.

Q10. Which algorithm is commonly used for Graph Coloring?

Ans: Backtracking and Branch & Bound.

Q11. Why is recursion used in Backtracking?

Ans: Recursion helps explore all possible solutions systematically.

Q12. What is pruning in Branch and Bound?

Ans: Pruning removes unnecessary branches that cannot give better solutions.

Q13. What is the state space tree?

Ans: A tree representing all possible solution states.

Q14. What is a feasible solution?

Ans: A solution that satisfies all constraints.

Q15. What is an optimal solution?

Ans: The best solution among all feasible solutions.

Q16. What is the time complexity of N-Queens?

Ans: $O(N!)$ in the worst case.

Q17. What is the advantage of Backtracking?

Ans: It reduces unnecessary computations by rejecting invalid solutions early.

Q18. What is the advantage of Branch and Bound?

Ans: It improves efficiency by pruning non-promising paths.

Q19. What is a dead state in Backtracking?

Ans: A state where no valid solution is possible.

Q20. What is a live node?

Ans: A node that can still produce a valid solution.

Q21. What is a safe state in N-Queens?

Ans: A position where a queen can be placed without conflict.

Q22. How do you check diagonal attacks in N-Queens?

Ans: Using row-column difference conditions.

Q23. Can two queens be placed in the same row?

Ans: No, it violates the constraints.

Q24. What is m-coloring problem?

Ans: Coloring a graph using at most m colors.

Q25. What is a constraint in Graph Coloring?

Ans: Adjacent vertices cannot have the same color.

Q26. What is the role of bounding function?

Ans: It determines whether a branch should be explored further.

Q27. What is DFS in Backtracking?

Ans: Depth First Search explores one solution path completely before backtracking.

Q28. What is the difference between Backtracking and Branch & Bound?

Ans: Backtracking checks constraints, while Branch & Bound also uses cost optimization.

Q29. What is combinatorial explosion?

Ans: Rapid growth of possible solutions with increasing input size.

Q30. Is N-Queens a CSP problem?

Ans: Yes, because it involves variables, domains, and constraints.

Q31. Is Graph Coloring an NP problem?

Ans: Yes, Graph Coloring is an NP-complete problem.

Q32. Give applications of Graph Coloring.

Ans: Scheduling, map coloring, and register allocation.

Q33. Give applications of N-Queens.

Ans: Artificial Intelligence and optimization problems.

Q34. Why are CSP problems important in AI?

Ans: Because many real-world AI problems involve satisfying constraints efficiently.

Chatbot Development

Q1. What is a chatbot?

Ans: A chatbot is a software application that interacts with users through text or voice conversations.

Q2. What is the purpose of a chatbot?

Ans: The purpose is to automate customer interaction and provide quick responses.

Q3. What is an elementary chatbot?

Ans: An elementary chatbot is a simple rule-based chatbot designed for basic conversations.

Q4. Which programming language is commonly used for chatbot development?

Ans: Python is commonly used for chatbot development.

Q5. What is Natural Language Processing (NLP)?

Ans: NLP is a technique that allows computers to understand human language.

Q6. What is a rule-based chatbot?

Ans: A chatbot that responds using predefined rules and conditions.

Q7. What is an AI-based chatbot?

Ans: A chatbot that learns and responds using machine learning and NLP.

Q8. What is pattern matching in chatbots?

Ans: Pattern matching identifies keywords or phrases in user input.

Q9. What is intent in chatbot systems?

Ans: Intent represents the purpose of the user query.

Q10. What is a response database?

Ans: A collection of predefined responses used by the chatbot.

Q11. What are the advantages of chatbots?

Ans: 24/7 availability, fast response, and reduced human effort.

Q12. What are the limitations of elementary chatbots?

Ans: They cannot understand complex or unexpected queries.

Q13. What is the difference between chatbot and virtual assistant?

Ans: Virtual assistants are more intelligent and support advanced tasks.

Q14. What is API in chatbot systems?

Ans: API allows communication between chatbot and external services.

Q15. What is machine learning in chatbots?

Ans: Machine learning helps chatbots improve responses using data.

Q16. What is context in chatbot conversation?

Ans: Context refers to previous conversation information.

Q17. What is a fallback response?

Ans: A default response when the chatbot cannot understand the query.

Q18. What are the applications of chatbots?

Ans: Customer support, banking, education, and e-commerce.

Q19. What is sentiment analysis in chatbots?

Ans: Detecting user emotions from text.

Q20. What is the difference between static and dynamic responses?

Ans: Static responses are fixed, while dynamic responses change based on input.

Q21. What are future trends in chatbot technology?

Ans: AI-powered assistants, voice bots, and multilingual chatbots.

Expert Systems

Q1. What is an Expert System?

Ans: An Expert System is an AI-based program that provides decisions similar to a human expert.

Q2. What is the main objective of an Expert System?

Ans: To solve complex problems using knowledge and rules.

Q3. What are the main components of an Expert System?

Ans: Knowledge Base, Inference Engine, and User Interface.

Q4. What is a Knowledge Base?

Ans: A database that stores facts and rules used by the system.

Q5. What is an Inference Engine?

Ans: The component that applies rules to generate decisions or suggestions.

Q6. What is the role of the User Interface?

Ans: It allows users to interact with the system.

Q7. What is rule-based reasoning?

Ans: Making decisions using predefined IF-THEN rules.

Q8. What is forward chaining?

Ans: A reasoning method that starts with facts and applies rules to reach conclusions.

Q9. What is backward chaining?

Ans: A reasoning method that starts with a goal and checks supporting facts.

Q10. What is a medical expert system?

Ans: A system that assists in disease diagnosis and medical recommendations.

Q11. Give an example of IF-THEN rule in hospital systems.

Ans: IF fever and cough exist THEN suggest flu test.

Q12. Give an example of IF-THEN rule in employee evaluation.

Ans: IF attendance is high and targets achieved THEN performance is excellent.

Q13. What is knowledge acquisition?

Ans: The process of collecting knowledge from experts.

Q14. What is a domain expert?

Ans: A person with specialized knowledge in a particular field.

Q15. What is the advantage of Expert Systems?

Ans: They provide fast and consistent decisions.

Q16. What is the limitation of Expert Systems?

Ans: They cannot think beyond programmed knowledge.

Q17. What is the difference between AI and Expert Systems?

Ans: Expert Systems are a branch of AI focused on decision-making.

Q18. What is explanation facility?

Ans: A feature that explains how the system reached a conclusion.

Q19. What are applications of medical expert systems?

Ans: Disease diagnosis, patient monitoring, and prescription support.

Q20. Can Expert Systems replace human experts completely?

Ans: No, they assist experts but cannot fully replace them.

Cloud Computing – Amazon EC2 & Google App Engine

Q1. What is Amazon EC2?

Ans: Amazon EC2 (Elastic Compute Cloud) is a web service that provides virtual servers in the cloud.

Q2. What is cloud computing?

Ans: Cloud computing is the delivery of computing services over the internet.

Q3. What is an EC2 instance?

Ans: An EC2 instance is a virtual machine running in AWS cloud.

Q4. What is AWS?

Ans: Amazon Web Services is a cloud platform providing computing, storage, and networking services.

Q5. What are the advantages of Amazon EC2?

Ans: Scalability, flexibility, reliability, and cost-effectiveness.

Q6. What is an AMI in EC2?

Ans: AMI (Amazon Machine Image) is a template used to create EC2 instances.

Q7. What is a Security Group in EC2?

Ans: A virtual firewall controlling inbound and outbound traffic.

Q8. What is EBS in AWS?

Ans: Elastic Block Store provides storage volumes for EC2.

Q9. What is Auto Scaling?

Ans: Automatically adjusting the number of EC2 instances based on demand.

Q10. What is Load Balancing?

Ans: Distributing traffic across multiple servers.

Q11. What is a region in AWS?

Ans: A geographical area containing AWS data centers.

Q12. What is an availability zone?

Ans: An isolated data center within an AWS region.

Q13. What is pay-as-you-go in AWS?

Ans: Users pay only for the resources they use.

Q14. What is Google App Engine?

Ans: Google App Engine is a cloud platform used to develop and host web applications.

Q15. What is Platform as a Service (PaaS)?

Ans: PaaS provides a platform for developing and deploying applications without managing hardware.

Q16. Which company developed Google App Engine?

Ans: Google developed Google App Engine.

Q17. What programming languages are supported by Google App Engine?

Ans: Python, Java, PHP, Go, Node.js, and others.

Q18. What is app.yaml file?

Ans: A configuration file used to define application settings.

Q19. What is automatic scaling in App Engine?

Ans: Automatically increasing or decreasing instances depending on load.

Q20. What command is used to deploy applications in App Engine?

Ans: gcloud app deploy

Q21. What is serverless computing?

Ans: Running applications without managing servers.

Q22. What is the difference between Google App Engine and Amazon EC2?

Ans: App Engine is PaaS, while Amazon EC2 is IaaS.

Q23. What is the advantage of Google App Engine?

Ans: Easy deployment and automatic scalability.

Q24. What are the applications of Google App Engine?

Ans: Website hosting, APIs, and enterprise applications.

Q25. What are the disadvantages of cloud computing?

Ans: Internet dependency and possible security concerns.

Salesforce – Apex & CRM Cloud

Q1. What is Salesforce?

Ans: Salesforce is a cloud-based Customer Relationship Management (CRM) platform.

Q2. What is CRM?

Ans: CRM stands for Customer Relationship Management used to manage customer data and interactions.

Q3. What is Apex in Salesforce?

Ans: Apex is an object-oriented programming language used in Salesforce.

Q4. Why is Apex used?

Ans: Apex is used to add custom business logic in Salesforce applications.

Q5. What is a custom object in Salesforce?

Ans: A user-created database table for storing specific data.

Q6. What is SOQL?

Ans: SOQL (Salesforce Object Query Language) is used to retrieve records from the Salesforce database.

Q7. What is SOSL?

Ans: SOSL (Salesforce Object Search Language) is used to search records across multiple objects.

Q8. What is a trigger in Apex?

Ans: A trigger is code that executes automatically before or after database events.

Q9. What are the events in Apex triggers?

Ans: Insert, Update, Delete, and Undelete events.

Q10. What is Visualforce?

Ans: A framework used to design custom user interfaces in Salesforce.

Q11. What is Lightning in Salesforce?

Ans: A modern framework for developing responsive Salesforce applications.

Q12. What is Governor Limit?

Ans: Limits imposed by Salesforce to ensure efficient resource usage.

Q13. What is multitenancy in Salesforce?

Ans: Multiple users share the same cloud infrastructure securely.

Q14. What is a sandbox in Salesforce?

Ans: A testing environment separate from the production environment.

Q15. What is a profile in Salesforce?

Ans: It controls user permissions and access.

Q16. What is DML in Apex?

Ans: Data Manipulation Language used for Insert, Update, Delete operations.

Q17. What are the DML operations?

Ans: Insert, Update, Delete, Upsert, Merge, and Undelete.

Q18. What is batch Apex?

Ans: Used to process large amounts of data asynchronously.

Q19. What is test class in Apex?

Ans: A class used to test Apex code functionality.

Q20. What is the minimum code coverage required in Salesforce?

Ans: 75%

Q21. What is the difference between Apex and Java?

Ans: Apex is cloud-based and runs on Salesforce servers, while Java is general-purpose.

Q22. What are applications of Salesforce?

Ans: Customer management, sales automation, marketing, and service support.

Q23. Why are Governor Limits important?

Ans: They ensure efficient resource sharing in the cloud.

Q24. What are the advantages of Salesforce?

Ans: Cloud access, scalability, security, and automation.

Q25. What are future trends in Salesforce Cloud?

Ans: AI integration, automation, and advanced cloud services.

Cloud Storage – HDFS & Distributed Systems

Q1. What is HDFS?

Ans: Hadoop Distributed File System is a distributed file storage system used in Hadoop.

Q2. What is Hadoop?

Ans: Apache Hadoop is an open-source framework for distributed storage and processing.

Q3. What is a cloud controller?

Ans: A cloud controller manages cloud resources and operations.

Q4. What is distributed storage?

Ans: Storing data across multiple systems or nodes.

Q5. What is a data block in HDFS?

Ans: A file is divided into smaller blocks for storage.

Q6. What is the default block size in HDFS?

Ans: Typically 128 MB.

Q7. What is a NameNode in HDFS?

Ans: The master node managing metadata and file information.

Q8. What is a DataNode in HDFS?

Ans: A node that stores actual data blocks.

Q9. What is file segmentation?

Ans: Dividing a file into smaller blocks or segments.

Q10. Why are files divided into blocks?

Ans: To improve storage efficiency and parallel processing.

Q11. What is replication in HDFS?

Ans: Creating multiple copies of data blocks for fault tolerance.

Q12. What is encryption?

Ans: Converting data into unreadable form for security.

Q13. Why is encryption used in cloud storage?

Ans: To protect files from unauthorized access.

Q14. What is decryption?

Ans: Converting encrypted data back into readable form.

Q15. What is fault tolerance in HDFS?

Ans: The ability to continue operation even if a node fails.

Q16. What is scalability in cloud computing?

Ans: Ability to increase resources according to demand.

Q17. What is metadata in HDFS?

Ans: Information about files and block locations.

Q18. What is authentication in cloud systems?

Ans: Verifying user identity before access.

Q19. What is authorization?

Ans: Granting permission to access resources.

Q20. What is client-server architecture?

Ans: A model where clients request services from servers.

Q21. What is data redundancy?

Ans: Storing duplicate copies of data for reliability.

Q22. What are applications of HDFS?

Ans: Big data analytics, cloud storage, and distributed computing.

Q23. What are the benefits of distributed cloud systems?

Ans: High availability, scalability, and reliability.

Q24. What is SaaS?

Ans: Software as a Service (SaaS) provides software applications through the internet.

Q25. What are the limitations of SaaS?

Ans: Internet dependency and security concerns.
